

Prediction-Synchronized Local Polynomial Smoothing

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lps_prediction_synchronized_design.tex

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Purpose

This design note defines the first version of prediction-synchronized local polynomial smoothing, abbreviated *PS-LPS*. The hyphenated name is used throughout this note because it remains readable in prose and parallels names such as S-LPL-TF. The goal is to keep the empirical strength and simplicity of local polynomial smoothing (LPS), while reducing the incoherence that can occur when many overlapping local polynomial charts are fitted independently.

The starting observation is that ordinary LPS constructs one local model around each anchor point. These models overlap heavily: many charts make predictions on the same data points. Ordinary LPS uses this overlap only indirectly. The proposed synchronized version uses it directly by penalizing disagreement between local predictions on shared overlap points.

The first implementation should be deliberately modest:

- no trend-filtering ℓ_1 penalty;
- no adaptive local parameter field yet;
- fixed candidate parameters (k, κ, p, d) inside each fit;
- one quadratic synchronization parameter $\lambda_{\text{sync}} \geq 0$;
- sparse least-squares implementation whenever possible.

This makes PS-LPS a natural bridge between ordinary LPS and S-LPL-TF / SLPLiFT. It is easier than full SLPLiFT because the objective is purely quadratic, but it tests the same core idea: local models should agree where their prediction domains overlap.

1 Ordinary LPS Notation

Let

$$x_1, \dots, x_n \in \mathbb{R}^D$$

be the observed covariate points, and let

$$y_i = f(x_i) + \varepsilon_i$$

be the observed response at point i . The ambient dimension is D , but the local chart dimension may be smaller.

For each anchor i , ordinary LPS chooses a support

$$N_i \subset \{1, \dots, n\}, \quad |N_i| = k,$$

usually the k nearest neighbors of x_i . A kernel κ assigns nonnegative weights

$$w_{ij} \geq 0, \quad j \in N_i.$$

The weight w_{ij} is large when x_j is close to anchor x_i , and is small when x_j lies near the edge of the local support.

At anchor i , construct local coordinates

$$z_{ij} \in \mathbb{R}^d, \quad j \in N_i,$$

for example by local PCA. Let

$$\phi_i(x_j) = \phi(z_{ij}) \in \mathbb{R}^q$$

be the local polynomial feature vector of degree p . For degree two, $\phi(z)$ contains the intercept, linear terms, squares, and cross terms. The number of columns is

$$q = 1 + d + \frac{d(d+1)}{2}.$$

Ordinary LPS fits a separate coefficient vector

$$\beta_i \in \mathbb{R}^q$$

at each anchor by weighted least squares:

$$\widehat{\beta}_i^{\text{LPS}} = \arg \min_{\beta_i} \frac{1}{2} \sum_{j \in N_i} w_{ij} \{y_j - \phi_i(x_j)^\top \beta_i\}^2.$$

The fitted value at the anchor is usually

$$\widehat{f}_i^{\text{LPS}} = \phi_i(x_i)^\top \widehat{\beta}_i^{\text{LPS}}.$$

Because each $\widehat{\beta}_i^{\text{LPS}}$ is fitted independently, neighboring charts may disagree even when they make predictions on the same overlap points.

2 Prediction Synchronization

PS-LPS keeps one local coefficient vector

$$\beta_i \in \mathbb{R}^q$$

for each anchor, but estimates all of them jointly. The local prediction made by chart i at point $r \in N_i$ is

$$g_i(x_r) = \phi_i(x_r)^\top \beta_i.$$

If another chart ℓ also contains x_r , then chart ℓ makes its own prediction

$$g_\ell(x_r) = \phi_\ell(x_r)^\top \beta_\ell.$$

The synchronization idea is to penalize

$$g_i(x_r) - g_\ell(x_r)$$

on shared overlap points.

Define the overlap between charts i and ℓ as

$$O_{i\ell} = N_i \cap N_\ell.$$

Only pairs with $O_{i\ell} \neq \emptyset$ can contribute synchronization rows. The basic quadratic synchronization penalty is

$$\mathcal{S}(\beta) = \frac{1}{2} \sum_{(i,\ell) \in \mathcal{E}_{\text{sync}}} \sum_{r \in O_{i\ell}} \omega_{i\ell r} \left\{ \phi_i(x_r)^\top \beta_i - \phi_\ell(x_r)^\top \beta_\ell \right\}^2.$$

Here $\mathcal{E}_{\text{sync}}$ is the set of chart pairs that are allowed to synchronize, and $\omega_{i\ell r} \geq 0$ is the overlap weight assigned to the shared prediction point r .

3 Recommended Overlap Weights

The natural first definition uses the product of the two chart weights:

$$\omega_{i\ell r}^{\text{raw}} = w_{ir} w_{\ell r}, \quad r \in O_{i\ell}.$$

This is attractive for exactly the reason discussed in the planning thread. A point x_r near the center of both charts receives a large product weight. A point near the boundary of one or both charts receives a small product weight. Thus the penalty asks charts to agree most strongly in the middle of their shared predictive region, and it is less strict near the edge of either local chart.

The raw product is conceptually clean, but its scale changes with the number of overlap points and with the kernel. For the first implementation, the recommended default is the pair-normalized product weight

$$\omega_{i\ell r} = |O_{i\ell}| \frac{w_{ir} w_{\ell r}}{\sum_{s \in O_{i\ell}} w_{is} w_{\ell s} + \epsilon}, \quad r \in O_{i\ell}.$$

The small constant $\epsilon > 0$ prevents division by zero. With this normalization,

$$\sum_{r \in O_{i\ell}} \omega_{i\ell r} \approx |O_{i\ell}|.$$

So the average overlap weight within a chart pair is approximately one, while the product structure still emphasizes the middle of the overlap.

This choice separates two roles:

- the relative weights inside one overlap are determined by the product $w_{ir} w_{\ell r}$;
- the global strength of synchronization is controlled by λ_{sync} , not by accidental kernel scale.

For auditability, the implementation should allow the raw product as an option, but the first experiments should use the normalized product as the default.

4 Objective Function

For fixed candidate parameters

$$\theta = (k, \kappa, p, d)$$

and synchronization strength λ_{sync} , define

$$\hat{\beta}(\theta, \lambda_{\text{sync}}) = \arg \min_{\beta_1, \dots, \beta_n} [\mathcal{L}_{\text{local}}(\beta) + \lambda_{\text{sync}} \mathcal{S}(\beta)],$$

where

$$\mathcal{L}_{\text{local}}(\beta) = \frac{1}{2} \sum_{i=1}^n \sum_{j \in N_i} w_{ij} \{y_j - \phi_i(x_j)^\top \beta_i\}^2.$$

The local loss says that each chart should fit the observed responses in its own support. The synchronization term says that overlapping charts should produce similar predictions on shared points.

When

$$\lambda_{\text{sync}} = 0,$$

the mathematical synchronization term is inactive. If no additional stabilization is used, the problem decomposes into ordinary independent LPS chart fits. If an explicit ridge stabilization is used, the $\lambda_{\text{sync}} = 0$ problem instead decomposes into independent ridge-stabilized LPS chart fits. The implementation and reports must state which case is being used. As λ_{sync} increases, the local models are increasingly forced to agree on overlaps. This is the key difference from ordinary LPS.

5 Ridge-Stabilized Implementation Contract

The first implementation now treats numerical ridge stabilization as an explicit modeling and reporting parameter rather than as an invisible solver detail. For a nonnegative ridge scale λ_{ridge} , the implemented objective is

$$\mathcal{L}_{\text{local}}(\beta) + \lambda_{\text{sync}} \mathcal{S}(\beta) + \frac{1}{2} \sum_{i=1}^n \rho_i \|\beta_i\|_2^2,$$

where ρ_i is a scale-relative ridge used for chart i . In the prototype, ρ_i is derived from the local weighted-design scale and the reported user parameter λ_{ridge} .

This gives two separate nesting checks:

- if $\lambda_{\text{ridge}} = 0$ and $\lambda_{\text{sync}} = 0$, PS-LPS should reproduce ordinary independent LPS fixed-chart fitted values;
- if $\lambda_{\text{ridge}} > 0$ and $\lambda_{\text{sync}} = 0$, PS-LPS should reproduce independent ridge-stabilized LPS fixed-chart fitted values.

Therefore ridge-stabilized PS-LPS must be compared against a matched ridge-stabilized LPS baseline before interpreting any remaining improvement as a synchronization effect.

6 Total Local GCV Inside PS-LPS

The earlier independent-LPS experiment showed that selecting global parameters by minimizing

$$\sum_{i=1}^n \text{GCV}_i$$

for independently fitted local charts is not a reliable parameter-selection rule on the first-batch examples. PS-LPS should nevertheless record the same kind of diagnostic, because the object being measured is now different.

After solving the synchronized problem, chart i has coefficients $\widehat{\beta}_i$. Its synchronized local support residual sum of squares is

$$\text{RSS}_i^{\text{PS}} = \sum_{j \in N_i} w_{ij} \{y_j - \phi_i(x_j)^\top \widehat{\beta}_i\}^2.$$

Let r_i be a local complexity proxy, initially the numerical rank of the weighted local design matrix for chart i . Define the synchronized local GCV diagnostic as

$$\text{GCV}_i^{\text{PS}} = \frac{\text{RSS}_i^{\text{PS}} / k}{(1 - r_i/k)^2}.$$

The total synchronized local GCV is

$$\text{GCV}_\Sigma^{\text{PS}} = \sum_{i=1}^n \text{GCV}_i^{\text{PS}}.$$

This is *not* the same as the total local GCV of ordinary LPS. The residuals are computed from prediction-synchronized charts, so they include the effect of overlap agreement. For this reason, the diagnostic may be more connected with cross-validation error than the independent-LPS version, even though it should still be treated as a diagnostic rather than as the primary selection rule in the first implementation.

7 Sparse Least-Squares Form

The objective is quadratic. Therefore the first implementation can be framed as one sparse least-squares problem.

Stack all local coefficients into

$$b = (\beta_1^\top, \dots, \beta_n^\top)^\top \in \mathbb{R}^{nq}.$$

Every local data-fit row has the form

$$\sqrt{w_{ij}} \phi_i(x_j)^\top \beta_i \approx \sqrt{w_{ij}} y_j.$$

This row touches only the block of variables belonging to chart i .

Every synchronization row has the form

$$\sqrt{\lambda_{\text{sync}} \omega_{i\ell r}} \left\{ \phi_i(x_r)^\top \beta_i - \phi_\ell(x_r)^\top \beta_\ell \right\} \approx 0.$$

This row touches only two blocks of variables: chart i and chart ℓ . Thus the global design matrix is sparse even though the number of local charts is n .

8 Prediction Rule

The primary fitted value should preserve ordinary LPS semantics:

$$\hat{f}_i = \phi_i(x_i)^\top \hat{\beta}_i.$$

That is, the prediction at x_i is read from the chart anchored at i . This makes direct comparison with ordinary LPS clean.

The implementation may also record an overlap-averaged diagnostic prediction

$$\hat{f}_i^{\text{avg}} = \frac{\sum_{\ell: i \in N_\ell} v_{\ell i} \phi_\ell(x_i)^\top \hat{\beta}_\ell}{\sum_{\ell: i \in N_\ell} v_{\ell i}},$$

where $v_{\ell i}$ can be the chart weight $w_{\ell i}$. This is useful diagnostically, but it should not be the primary fitted value in the first experiment because it changes two things at once: synchronization and prediction aggregation.

9 Cross-Validation Without Response Leakage

For cross-validation, validation responses must not enter the local data-fit loss. If V is a validation fold, then data-fit rows with response index

$$j \in V$$

should receive zero response weight in $\mathcal{L}_{\text{local}}$. The synchronization rows may still include validation covariate locations because they use only X , chart bases, and kernel weights. This does not leak validation responses.

The held-out prediction at x_j , $j \in V$, is computed after solving the fold-specific synchronized least-squares problem. The CV score is then the usual held-out RMSE over validation responses. The candidate grid should include

$$\lambda_{\text{sync}} \in \{0, \lambda_1, \lambda_2, \dots\},$$

so the ordinary LPS case is always present as a nested reference.

10 First Implementation Defaults

The first implementation should be intentionally narrow.

Object	First default	Reason
Local chart	ordinary local PCA	matches current LPS path
Polynomial degree	degree 2	current preferred LPS setting
Chart dimension	fixed d , plus later <code>auto/local.auto</code>	simplest solver contract
Overlap pairs	nearby anchor pairs with nonempty overlap	avoids all-pairs blowup
Overlap weight	normalized product $w_{ir}w_{\ell r}$	emphasizes overlap interior
Prediction	anchor prediction $\phi_i(x_i)^\top \beta_i$	comparable to LPS
Solver	sparse QR/least-squares path	objective is quadratic

The first validation should use the frozen first-batch non-manifold datasets where ordinary LPS `auto` and `local.auto` results already exist. This keeps the first comparison focused:

ordinary LPS versus prediction-synchronized LPS.

11 Diagnostics

The first experiment should report:

- selected λ_{sync} , support size, kernel, degree, and chart dimension;
- Truth RMSE, Observed RMSE, and CV RMSE where synthetic truth is available;
- total synchronized local GCV

$$\text{GCV}_{\Sigma}^{\text{PS}} = \sum_i \text{GCV}_i^{\text{PS}};$$

- synchronization energy

$$\mathcal{S}(\hat{\beta});$$

- local prediction disagreement before and after synchronization;
- whether pointwise Truth-RMSE contribution maps improve in regions where `local.auto` previously lost to `auto`;
- runtime and sparse-system size diagnostics.

The most important scientific question is not whether synchronization always improves the average score on the first attempt. The first question is whether the method changes the right object: does it reduce incoherent overlap predictions without collapsing into oversmoothing or interpolation?

12 Out of Scope for the First Version

Several extensions are attractive but should not be mixed into the first implementation:

- an ℓ_1 trend-filtering penalty;
- locally varying support size, kernel, degree, or chart dimension;
- GCV-driven adaptive local parameter fields;
- overlap-averaged prediction as the main estimator;
- second-order local SVD charts;
- theoretical rate claims.

These can be added later if the quadratic synchronized LPS core behaves sensibly.

13 Summary

Prediction-synchronized LPS estimates all local polynomial charts jointly. The new ingredient is the quadratic penalty

$$\sum_{(i,\ell)} \sum_{r \in O_{i\ell}} \omega_{i\ell r} \left\{ \phi_i(x_r)^\top \beta_i - \phi_\ell(x_r)^\top \beta_\ell \right\}^2.$$

The recommended first overlap weight is the normalized product of the two ordinary chart weights. This keeps the middle-of-overlap intuition while making the scale of the synchronization parameter easier to compare across datasets and candidate settings.

The immediate next step is an implementation phase that builds the sparse quadratic system for fixed $(k, \kappa, p, d, \lambda_{\text{sync}}, \lambda_{\text{ridge}})$, verifies the two lambda-zero nesting checks above, and then runs a small grid over λ_{sync} on the frozen first-batch datasets with ordinary LPS and ridge-matched LPS baselines kept separate.